

Scalability & Future Plan

Date: 29 September 2026 **Team ID:** SWTID-2026-4886
Project: AnswerBot – AI-powered FAQ and Customer Support Automation

Current Scalability

- Stateless JWT-based API allows multiple server instances with shared database access.
- MongoDB Atlas can support managed database scaling as usage grows.
- Routes and services are modular, allowing independent optimization.
- External AI integration is isolated from core FAQ persistence.

Future Plan

- Deploy behind a reverse proxy/load balancer.
- Add Redis caching for frequently requested FAQs.
- Introduce vector embeddings and semantic retrieval for stronger context-aware answers.
- Add rate limiting, API monitoring and structured logs.
- Add role-based administration and moderation.
- Add automated unit/integration/load tests.
- Add frontend dashboard and analytics.

Risks / Controls

Risk	Mitigation
AI incorrect response	Use FAQ knowledge/context, validation and human review for published content.
API key exposure	Use environment variables and never commit secrets.
High AI usage cost	Rate limits, caching and controlled prompt size.
Database load	Indexes, pagination and managed MongoDB scaling.